

MARWAN MAHMOUD ELSAYED

Robotics Engineer
Cairo, Egypt

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PROFESSIONAL SUMMARY

Robotics-focused Mechatronics Engineering student with hands-on experience in embedded systems, electronics, and hardware-software integration through engineering projects and the DEPI AI & Machine Learning Diploma. Proven ability to design, build, and test autonomous robotic systems, including mobile robots and quadruped platforms. Demonstrates strong problem-solving skills, technical leadership, and experience working in collaborative engineering environments.

EDUCATION

- Nahda University**, Beni Suef, Egypt

Mechatronic Systems Engineering

(2024 – Expected 2027)

- October University for Modern Sciences and Arts**, 6th October City, Egypt

Mechatronic Systems Engineering

(2020 – 2024)

SKILLS

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|---|--------------------------------|
| • CAD Software (AutoCAD, SOLIDWORKS) | • Digital & Analog Electronics |
| • Programming Languages (C, C++, Python) | • Critical Thinking |
| • Embedded Systems & Robotics (Arduino, Sensors, Motor Drivers) | • Problem Solving |
| • Collaboration Tools (GitHub, Notion) | • Effective Communication |
| • Control Concepts (PID, Basic Control Logic) | • Project Management |

PROJECTS

- Arduino Line-Following Robot**
Designed, built, and programmed an autonomous line-following robot using IR sensors, L298N motor driver, and PID-based control logic, achieving approximately 95% tracking accuracy at 0.8 m/s. Implemented optimized C++ firmware and validated performance across multiple track layouts.
- Bluetooth Controlled Quadruped Spider Robot**
Designed and implemented a quadruped robotic platform featuring coordinated servo-driven leg movement. Diagnosed and fixed control-timing delays that caused unstable high-speed actuation, enabling smooth, stable, and natural locomotion through optimized gait sequencing and embedded control logic.

EXPERIENCE

- **IEEE – Nahda University Branch** *(Nov 2025- Present)*
Technical Member | Robotics & Programming Focus
 - Contributed to developing educational technical content in Python programming and GitHub version control.
 - Supported hands-on programming workshops, helping students understand sensors, logic flow, and basic automation concepts.
 - Assisted in technical documentation, learning material design, and peer mentoring for engineering students.
 - Collaborated with IEEE team members to improve technical engagement and learning outcomes in engineering-related activities.
- **Digital Egypt Pioneers Initiative (DEPI)** *(Dec 2025- Present)*
Intern | AI & Machine Learning
 - Participated in an intensive AI & Machine Learning training program focused on real-world engineering applications.
 - Worked on intelligent systems and robotics-related problem solving, integrating Python, machine learning models, and automation logic.
 - Contributed to system planning, task coordination, model testing, and technical integration within a multidisciplinary engineering team.
 - Supported debugging, optimization, and documentation of project pipelines to improve system performance and clarity.
 - Gained practical experience in data processing, AI workflows, technical reporting, and collaborative engineering development.

CERTIFICATIONS

- **ALX AiCE - AI Career Essentials** | ALX Africa
8-week program in AI Augmented Professional Development skills *(July 2024)*
- **Introduction to Python Training** | AITB Educational Services
20 Training Hours in Python Programming Essentials *(April 2024 - May 2024)*

LANGUAGES

Arabic: Native

English: Fluent

Spanish: Beginner

